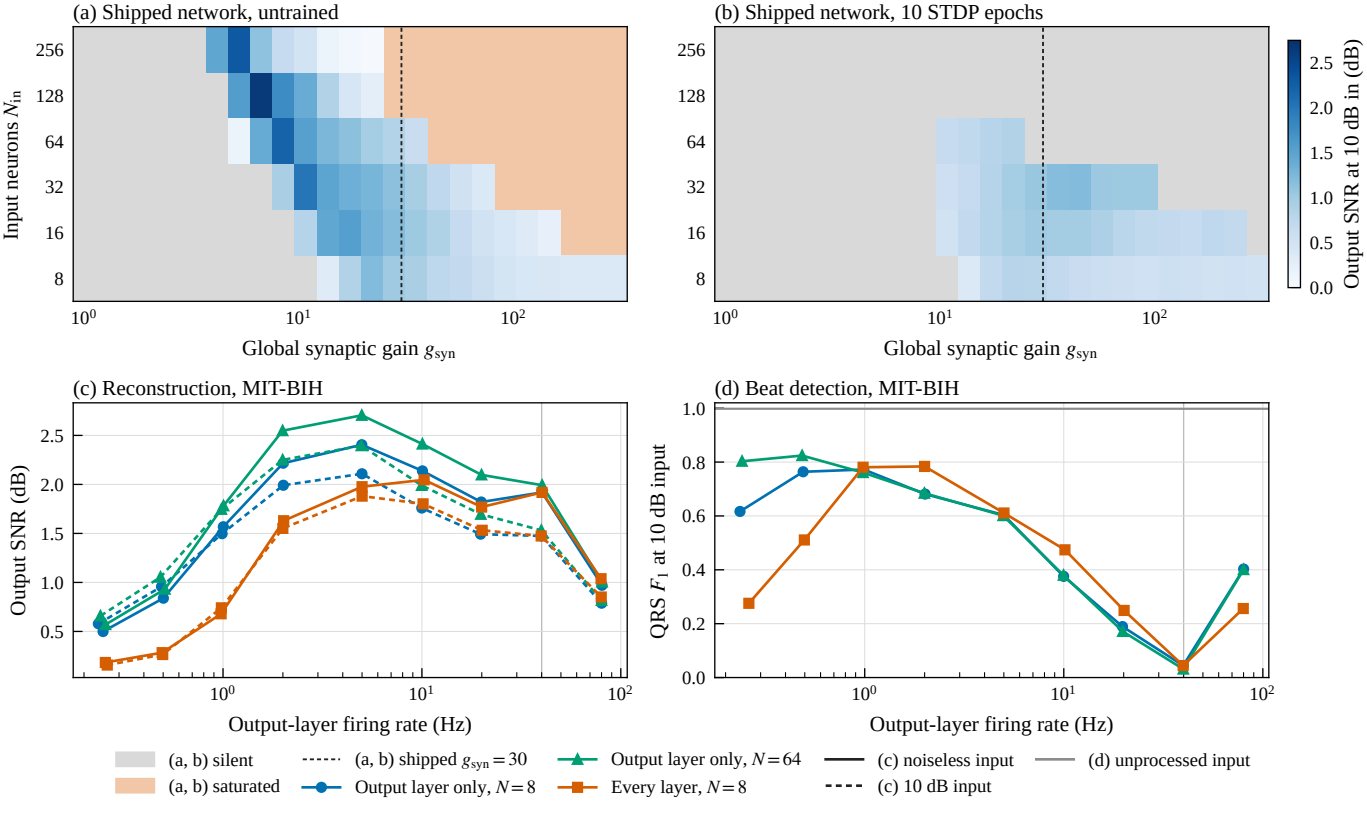
(a) Shipped network, untrained
(b) Shipped network, 10 STDP epochs
Input neurons N_{in}
256
128
64
32
16
8
10^0
10^1
10^2
Global synaptic gain g_{syn}
Output SNR at 10 dB in (dB)
2.5
2.0
1.5
1.0
0.5
0.0
(c) Reconstruction, MIT-BIH
(d) Beat detection, MIT-BIH
Output SNR (dB)
QRS F_1 at 10 dB input
Output-layer firing rate (Hz)
(a, b) silent
(a, b) saturated
(a, b) shipped $g_{\text{syn}} = 30$
Output layer only, $N = 8$
Output layer only, $N = 64$
Every layer, $N = 8$
(c) noiseless input
(c) 10 dB input
(d) unprocessed input